A THERMAL IMAGING-BASED APPROACH FOR FIRE DETECTION IN A CONTROLLED AND OPEN SPACE ENVIRONMENT WITH MOBILE ALERTING SYSTEM

A Thesis Presented to the Faculty of
Computer Engineering Department
College of Engineering
University of St. La Salle
Bacolod City

In Partial Fulfillment
of the Requirements for the Degree
Bachelor of Science in Computer Engineering

ODSEY BANDOJO
VHIERON BAREZA
REYMART LOUIE CAPAPAS

March 2026

APPROVAL SHEET

The research entitled **A THERMAL IMAGED-BASED APPROACH FOR FIRE DETECTION IN A CONTROLLED AND OPEN SPACE ENVIRONMENT WITH MOBILE ALERTING SYSTEM** presented by **ODSEY BANDOJO**, **VHIERON BAREZA**, and **REYMART LOUIE CAPAPAS** in partial fulfillment of the requirements for Bachelor of Science in Computer Engineering of the University of St. La Salle has been evaluated and approved by the panel of evaluators.

PANEL OF EVALUATORS

CHARINA D. PUENTEVELLA, MIT
Chair Panelist

CHRISTIAN SEGUNLA
Panelist

MILAGROS S. ORACA, LPT. PhD
Panelist

JASON QUINTANILLA
Technical Adviser

JEFFREY FUENTES
Thesis Adviser

ACKNOWLEDGEMENT

This research and project would not have been possible without the incredible support and contributions of many individuals.

We would like to sincerely thank Charina D. Puentevella, MIT, for her guidance, encouragement, and unwavering support throughout this study.

We are also deeply grateful to Jason Quintanilla for his time, patience, and expertise as our technical adviser.

We would also like to extend our gratitude to our panelists Dr. Mila Oraca and Mr. Christian Segunla for their guidance and recommendation for the improvement of our system.

We are also immensely thankful to the University of St. La Salle for providing us with the resources and facilities, necessary to make this study possible.

Finally, we extend our deepest gratitude to our families and friends for their constant encouragement and support. This project would not have been possible without the collective help and dedication of everyone involved, and we truly appreciate all the efforts that made this project a success.

INTRODUCTION

Background of the Study

Existing research has explored the integration of imaging and sensor technologies for improved fire detection. Sousa et al. (2020) demonstrated that thermal infrared sensing maintains reliability even under conditions of poor visibility, where smoke and low light typically reduce the accuracy of visual cameras. Their study confirmed that thermal infrared imaging could detect fire signatures earlier than conventional RGB-based systems. The research gap lies in the lack of lightweight, community-scale applications of thermal infrared sensing that do not depend on high-bandwidth infrastructure.

Similarly, Vuong et al. (2024) developed an experimental fire detection and alarm system that integrates IoT-based environmental sensing with real-time image analysis. Their findings showed that combining image processing with IoT sensors reduced false positives compared to threshold-based systems alone, thus improving reliability in fire detection. Nonetheless, their system required relatively advanced computing hardware, making it less suited for rural communities with limited power and network access. The research gap here is the absence of simplified, edge-deployable versions of IoT–image processing hybrids that could operate in areas with constrained resources.

Given these limitations, there is a clear need for a lightweight, edge-based fire detection and alert system designed specifically for rural barangays. By processing data locally through embedded systems and delivering alerts via mobile push notifications and SMS, such an approach reduces dependence on cloud infrastructure while ensuring timely communication of fire risks. This study therefore seeks to address the gaps identified in existing literature by

adapting proven concepts of thermal imaging and IoT integration into a community-level early warning tool.

Fire incidents in the Philippines remain a persistent challenge, particularly in rural barangays. According to the Bureau of Fire Protection (BFP, 2023), there were 16,426 recorded fire incidents nationwide, representing a 24.5% increase from the previous year. Many of these incidents resulted in the destruction of property, agricultural losses, and risks to human life. In remote communities, where monitoring systems and response capacity are limited, fires are often detected too late, underscoring the urgent need for reliable early detection and alerting mechanisms that can function even in low-resource environments.

Statement of the Problem

In rural communities, fires often begin unnoticed in unguarded spaces whether in open and controlled environments, or near homes due to the absence of continuous monitoring systems. While local residents may respond when they see fire, the reality is that many incidents start where no one is present. This study addresses key challenges that prevent early, community-led action in rural fire emergencies by asking the following questions:

1. What is the performance of the automated fire detection system in a controlled environment in terms of:
    a. detection rate
    b. detect heat signatures
    c. detect gas & smoke

   d. notification speed

2. What is the performance of the automated fire detection system in open space in terms:
   a. detection rate
   b. detect heat signatures
   c. detect gas & smoke
   d. detect ambient temperature & humidity
   e. notification speed
3. What is the performance of the automated fire detection system when there is an actual fire in terms of:
   a. detection rate
   b. detect heat signatures
   c. detect gas & smoke
   d. detect ambient temperature & humidity
   e. notification speed
4. Is there a significant difference in the performance of the automated fire detection system in a controlled environment, open space and when there is an actual fire in terms of:
   a. detection rate
   b. detect heat signatures
   c. detect gas & smoke
   d. detect ambient temperature & humidity
   e. notification speed

Conceptual Framework

This study adopts an Input–Process–Output (IPO) model to describe how the proposed system detects and communicates fire risks in rural communities.

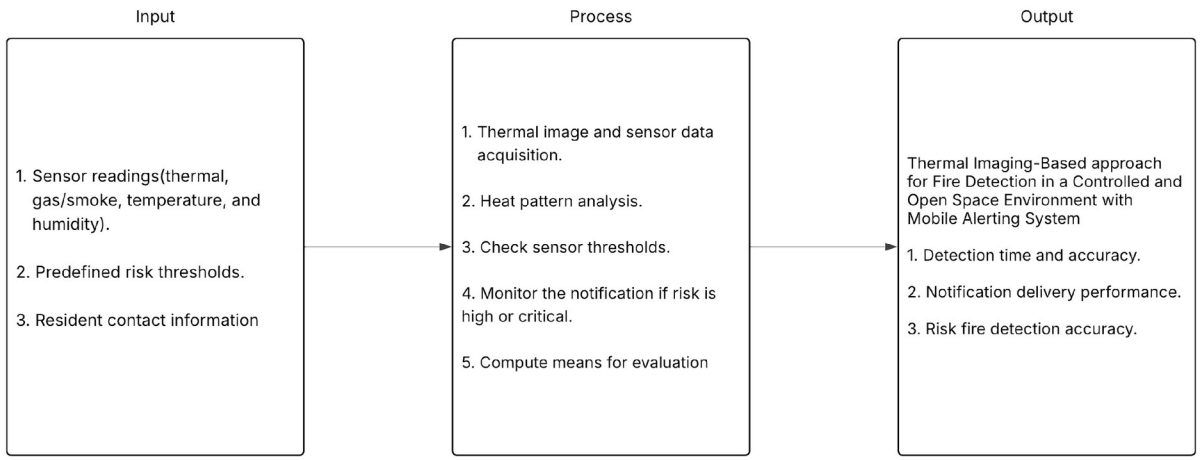


Figure 1. *Conceptual Framework of the Study*

Scope and Limitations

This study is limited to the design, development, and testing of a community-level fire risk alert system for rural barangays. It focuses on detecting fire-prone conditions within small scale, barangay-managed areas such as Kabankalan City, Brgy. Tapi's open and controlled areas with residences. The system is designed for local deployment and does not aim to cover large-scale, nationwide, or regional fire monitoring. The evaluation of the study is confined to measuring detection rate, heat signatures, gas & smoke, ambient temperature & humidity, and notification speed.

The scope of this study is constrained by several factors. The system is intended only as an early warning and monitoring tool; it does not include automated firefighting or suppression capabilities. Its deployment is dependent on the availability of electricity and at least minimal mobile network coverage, as alerts cannot be delivered without these. The study does not address integration with national fire monitoring agencies or large-scale disaster management systems. Furthermore, testing and validation are limited to simulated or small-scale field environments, which may not capture the full complexity of large fire events.

Significance of the Study

The development of a community-level fire risk alert system using edge image classifier, thermal sensing, and mobile-based notifications is significant to the following:

Barangay Community. The system is designed to benefit entire barangay communities, including both residents and local officials. Residents gain timely awareness through push notifications and SMS, enabling them to act quickly and reduce the risks of property loss, injury, or delayed evacuation. At the same time, barangay captains and tanods can use the alerts to monitor fire-prone areas more effectively, coordinate community responses, and mobilize local firefighting measures without relying solely on outside reports.

Local Government Units (LGUs). This system provides LGUs with a tool to strengthen their disaster risk reduction and management (DRRM) capabilities. It enables early warnings even in remote areas where traditional fire monitoring is lacking, allowing LGUs to respond proactively without relying on human reporting alone. The system also supports the goals of SDG 11: Sustainable Cities and Communities.

Farmers. Farms and open lands are highly vulnerable to fast-spreading fires, especially when no one is around to detect them early. This system helps farmers by providing timely alerts via mobile notifications, allowing them to act quickly to protect crops, equipment, and property before fires escalate.

Future Researchers. This study contributes to the body of research on community-based safety systems that use edge computing and thermal imaging. Future researchers may expand on this work by improving the detection accuracy, incorporating automated suppression mechanisms, or adapting it to other disaster scenarios.

Definition of Terms

This study involves several technical terms related to fire risk detection, embedded systems, and mobile-based alerting, particularly in rural settings.

Thermal Camera. Used to capture heat patterns in the environment up to 7m. In this system, the MLX90640 thermal camera detects unusually hot areas that could indicate early signs of fire.

Gas and Smoke Sensor. The MQ2 sensor detects flammable gases or smoke in the air. This supports the system's decision-making by helping confirm if fire-like conditions are present.

Temperature and Humidity Sensor. The DHT22 measures ambient temperature and humidity levels with ±0.5°C and ±2% RH for humidity. These readings help determine if conditions are dry and hot enough to increase fire risk.

Raspberry Pi Zero 2 W. Acts as the main processing unit for the system. It collects and analyzes sensor data on-site and runs the TinyML model to decide when to send fire risk alerts.

TinyML. A lightweight machine learning approach designed to run on small devices like the Raspberry Pi Zero 2 W. In this system, it processes thermal images and sensor data to identify fire-like patterns and improve accuracy over simple threshold-based detection.

Fire Risk Level. Describes how likely a fire is to occur. The system categorizes risk as High or Critical based on TinyML analysis of sensor values and heat patterns.

Push Notification. A real-time alert sent to a user's smartphone through a mobile app. It notifies residents when fire risk is high, enabling early action.

SMS Alert. A backup method of communication. If mobile data is unavailable, the system uses SMS to send fire risk alerts to ensure wider coverage.

SMS Gateway. A centralized alert broadcasting system located at the barangay hall. It consists of an Arduino Mega 2560 microcontroller with built-in WiFi and multiple GSM modules that work together to send mass SMS alerts to all registered residents within seconds of fire detection.

Arduino Mega 2560. A microcontroller board that serves as the main processor for the SMS Gateway system. It monitors the cloud database for new fire events and coordinates the GSM modules to broadcast alerts to the community.

GSM Module. A cellular communication device (SIM800L) that enables the system to send SMS messages through the local mobile network. Multiple GSM modules work in parallel to rapidly broadcast alerts to hundreds of residents simultaneously.

Mobile Application. A React Native-based app connected to Supabase. It receives alerts, displays current fire risk levels, and keeps communities informed in real-time.

Supabase. A cloud-based database platform that serves as the central communication hub between sensor nodes and the SMS Gateway. It stores fire event data, manages user information, and enables real-time synchronization across all system components.

Review of Related Literature

This section presents the relevant concepts and studies that are related to the current study.

Edge Computing for Real-Time Fire Detection

Edge computing has shown significant potential in improving real-time fire detection by enabling low-latency, localized processing and reducing cloud dependency. Mahdi and Mahmood (2022) reported up to 99% accuracy with a YOLOv5-based framework but noted sensitivity issues in large-area fires and deployment challenges in remote zones. Guo et al. (2022) integrated edge devices with cloud IoT platforms to balance processing and analytics, though they faced trade-offs in managing heterogeneous sensor data. Su et al. (2023) embedded edge capabilities into PLCs, enhancing responsiveness yet encountering scalability and compatibility issues with newer IoT standards. Wenzheng and Jie (2023) improved YOLOv7 for edge use, boosting detection precision but constrained by hardware limitations, while Vazquez et al. (2025) designed a lightweight YOLOv5 variant for resource-limited environments, with some performance loss under smoke and poor lighting. Mu et al. (2025) also demonstrated UAV-mounted edge processing for rapid detection but faced restrictions from flight duration, weather, and regulations. Collectively, these works emphasize that while edge computing enhances

detection accuracy and speed, challenges in hardware capacity, environmental conditions, and system scalability remain.

Integration of IoT with Fire Detection Systems

IoT-based fire detection systems leverage interconnected sensors and communication networks to provide continuous monitoring, rapid alerts, and remote accessibility. Komalapati et al. (2021) developed a smart IoT-enabled surveillance system that demonstrated effective real-time detection, while Mekni (2022) proposed a low-cost, scalable IoT framework suitable for widespread deployment. Vuong et al. (2024) enhanced detection accuracy by combining IoT sensor data with image processing, and Das et al. (2022) utilized IoT-enabled cameras to identify fire indicators such as smoke and flames. Guo et al. (2022) further integrated IoT with edge and cloud platforms, improving responsiveness and scalability through balanced local and centralized processing. In spite of these advances, challenges remain, including reliability issues from network latency and power interruptions, interoperability among heterogeneous devices, cybersecurity and privacy concerns, and susceptibility to false alarms caused by environmental noise. Cost and maintenance demand also hinder large-scale adoption, highlighting the need for solutions that strengthen robustness, security, and efficiency for long-term IoT-enabled fire safety systems.

Deep Learning and Computer Vision in Fire Detection

Recent studies highlight the effectiveness of deep learning and computer vision in improving fire detection accuracy and adaptability across diverse environments. Almeida et al. (2022) developed EdgeFireSmoke, a lightweight CNN optimized for edge deployment, which achieved efficient real-time smoke and fire detection but was constrained by limited scalability for large-scale monitoring. Mohammed (2022) and Yang et al. (2024) applied CNN-based

models that demonstrated high precision in outdoor and urban scenarios, though their performance was affected by environmental factors such as lighting variations and occlusions. Tsai et al. (2023) employed YOLOv4 with thermal imaging to detect humans in smoke-heavy conditions, enhancing rescue operations but requiring high-quality thermal sensors, which may increase costs. Similarly, Wenzheng and Jie (2023) improved YOLOv7 with attention mechanisms for wildfire detection, boosting accuracy but facing challenges with computational demand on edge hardware. Vazquez et al. (2025) further optimized YOLOv5 for resource-constrained devices, balancing speed and accuracy but with reduced robustness in smoke-dense or low-light conditions. Taken together, these studies demonstrate that deep learning models significantly advance detection precision, speed, and situational awareness, yet limitations related to hardware requirements, environmental variability, and scalability remain barriers to broader implementation.

Hybrid Systems: IoT + Edge + Deep Learning

Recent research on hybrid fire detection frameworks addresses the challenge of achieving real-time monitoring while managing latency, bandwidth, and computational efficiency. Dhir et al. (2023) developed a system integrating IoT sensor networks, edge computing, and deep learning, which enabled timely wildfire detection and prediction but required careful coordination between distributed nodes. Yang et al. (2024) proposed an edge–cloud hybrid architecture for image-based fire detection that reduced latency through edge processing while leveraging cloud scalability, though reliance on stable connectivity remained a limitation. Guo et al. (2022) combined edge devices with cloud platforms to optimize bandwidth usage and improve system responsiveness, yet scalability issues arose in high-density sensor deployments. Mu et al. (2025) further advanced hybrid designs by balancing computational loads between

edge and cloud, enhancing adaptability but introducing complexity in system management. Collectively, these studies show that hybrid approaches improve scalability, adaptability, and robustness in fire detection, while persistent challenges include system coordination, network stability, and deployment complexity.

Sensor Network-Based Detection with Machine Learning

Research on wireless sensor networks (WSNs) combined with machine learning has focused on improving reliability and reducing false alarms in fire detection. Ateeq and Momani (2020) integrated WSNs with image processing to detect early fire indicators in remote areas, achieving timely alerts but facing challenges with energy consumption and network stability. Dampage et al. (2022) applied supervised machine learning algorithms to WSN data, which improved detection accuracy and minimized false positives, though their approach required extensive training datasets for optimal performance. Ferreira et al. (2020) developed an autonomous system combining environmental sensors, computer vision, and automated response mechanisms, enabling early wildfire detection and control but increasing system complexity and maintenance requirements. Taken together, these studies highlight that coupling sensor networks with intelligent algorithms enhances accuracy, scalability, and responsiveness, while practical issues such as energy efficiency, data requirements, and system complexity remain significant barriers.

Synthesis

Recent literature highlights significant advancements in smart fire detection systems through the integration of edge computing, IoT, and deep learning technologies. Edge computing notably improves fire detection speed and reliability by enabling localized, low-latency data processing and decision-making. Studies by Mahdi and Mahmood (2022), Guo et al. (2022), Su et al. (2023), and Vazquez et al. (2025) demonstrate how edge-based frameworks using lightweight deep learning models like enhanced YOLO variants (YOLOv5 and YOLOv7) can achieve near real-time wildfire and industrial fire detection with high accuracy while overcoming computational and bandwidth constraints typical of cloud-only solutions. The use of UAVs with thermal and color imaging processed at the edge further enhances rapid forest fire detection capabilities (Mu et al., 2025).

IoT integration complements these advances by providing interconnected sensors for continuous, remote fire monitoring and rapid alerting. Systems developed by Komalapati et al. (2021), Mekni (2022), Vuong et al. (2024), and Das et al. (2022) showcase scalable, low-cost IoT architectures that fuse sensor data with image processing for improved situational awareness. The combination of IoT with edge computing and cloud platforms, as demonstrated by Guo et al. (2022), enables balanced processing between local responsiveness and cloud-based analytics.

Deep learning approaches, specifically convolutional neural networks optimized for edge deployment such as EdgeFireSmoke (Almeida et al., 2022), YOLO models improved with attention mechanisms (Wenzheng and Jie, 2023), and lightweight customized CNNs (Mohammed, 2022; Yang et al., 2024), significantly enhance the accuracy and speed of visual fire detection. These models are competent in complex environments, including smoke-heavy urban and natural settings.

Hybrid systems integrating IoT, edge computing, and deep learning (Dhir et al., 2023; Yang et al., 2024; Guo et al., 2022; Mu et al., 2025) offer scalable, adaptive, and robust frameworks that leverage strengths of each technology layer to maximize detection efficacy and response times.

Complementing vision-based approaches, wireless sensor networks combined with machine learning techniques show promise in improving early fire detection and reducing false alarms (Ateeq and Momani, 2020; Dampage et al., 2022; Ferreira et al., 2020). These systems fuse environmental sensor data with intelligent algorithms, enabling scalable, real-time responsiveness in remote or challenging environments.

METHODS AND MATERIALS

Research Design

This study employs a developmental–experimental research design, integrating the iterative development of a prototype system with controlled performance testing. A recent example of such an approach is Palmerola (2024), who applied a developmental research design to evaluate an e-learning system through iterative cycles of planning, implementation, and refinement.

The developmental component supports iterative design improvements, essential for tailoring your system to the specific needs of rural barangays, while the experimental component allows for empirical performance evaluation, such as detection accuracy and responsiveness. This blend is especially suitable for technology-based innovation projects like your fire detection and alert system, where both product development and validation are critical.

System Architecture

The proposed system architecture is designed to provide localized, real-time fire risk monitoring and community-level alerting without full reliance on cloud infrastructure. At its core, the architecture integrates a thermal imaging sensor, environmental sensors, and a lightweight embedded processor (Raspberry Pi Zero 2 W) to process data directly at the edge.

Sensor nodes continuously capture thermal images, gas concentration, and environmental readings. These inputs are analyzed on-node using a lightweight rule-based logic combined with TinyML for identifying fire-like heat patterns. When thresholds indicating high fire risk are met,

the system triggers the alerting module. Alerts are transmitted via two channels: push notifications through a mobile application and SMS messaging through a GSM Router.

The mobile application, developed with React Native and connected to Supabase, serves as the community interface, displaying risk levels and alert logs. By combining edge processing with mobile-based dissemination, the architecture ensures timely awareness even in areas with unstable internet access.

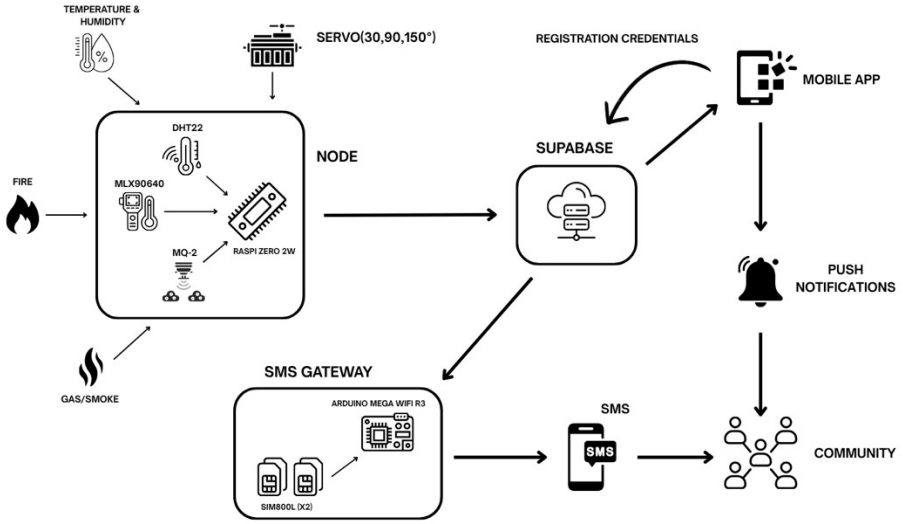


Figure 2. *System Architecture*

Context Level Diagram

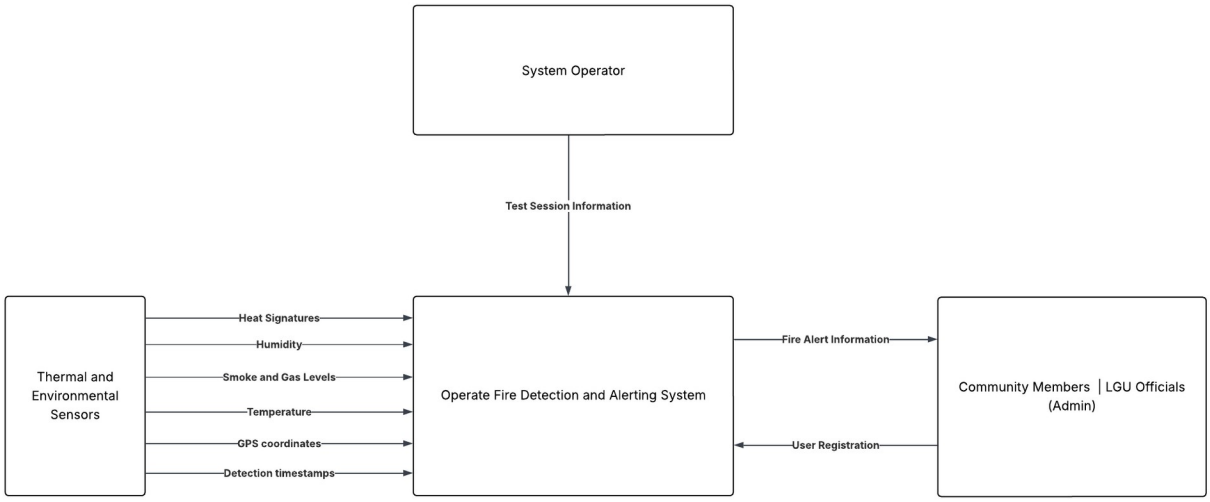


Figure 3. *Context Level Diagram*

Data-Flow Diagram – Level 1

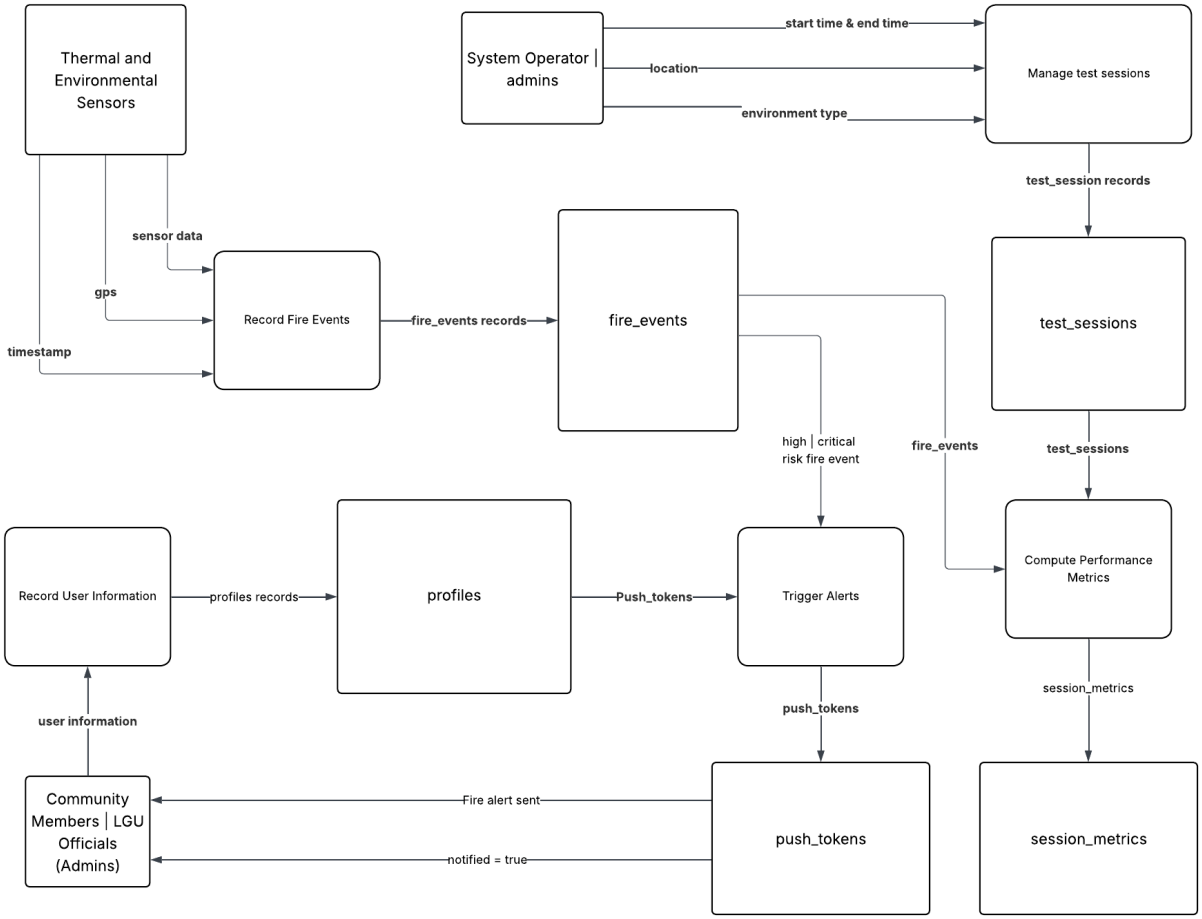


Figure 4. *Data-Flow Diagram - Level 1*

Entity Relationship Diagram

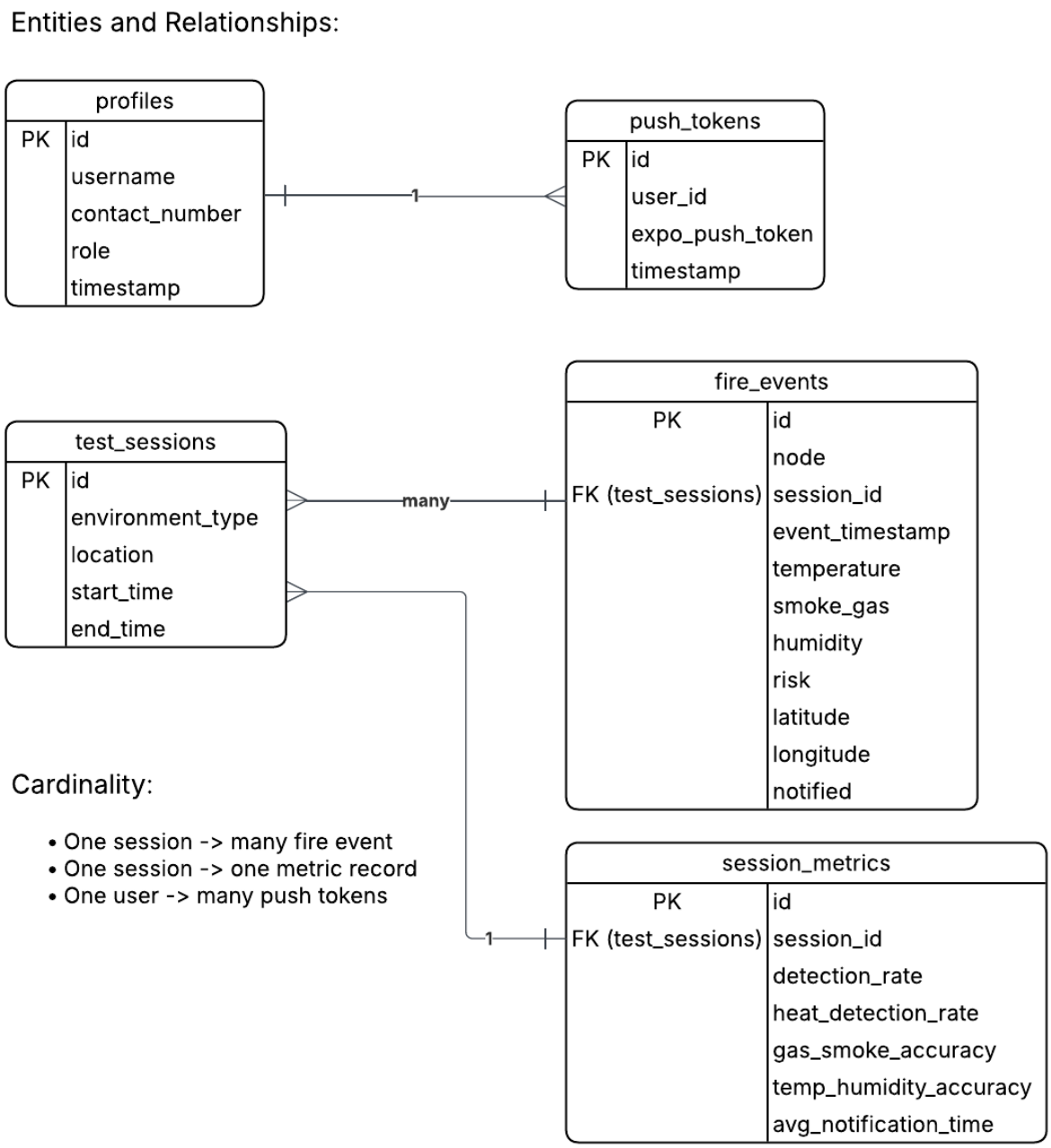


Figure 5. *Entity Relationship Diagram*

Database Schema

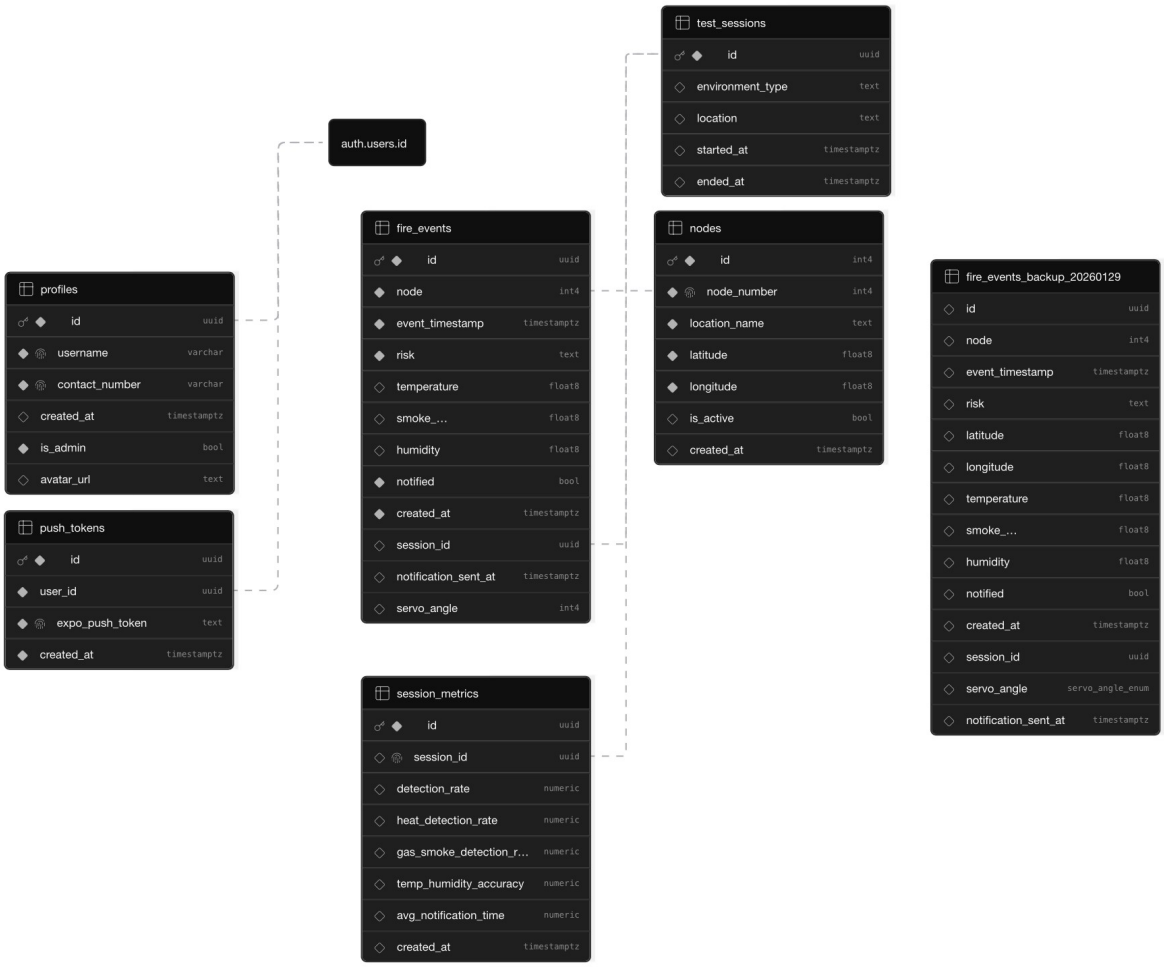


Figure 6. *Database Schema*

Data Dictionary

Table Name	Attributes (Fields)	Key (PK/FK)
profiles	*id* *username* *contact_number* *role (community | admin)* *timestamp*	id (PK), references auth.users.id
fire_events	id (PK) session_id (could be NULL) node event_timestamp temperature smoke_gas humidity risk latitude longitude notified	id (PK) session_id (FK → test_sessions.id)
push_tokens	id (PK) user_id expo_push_token timestamp	id (PK), user_id (FK)
test_sessions	id (PK) environment_type location start_time end_time	id (PK)
session_metrics	id (PK) session_id detection_rate heat_detection_rate gas_smoke_detection_rate temp_humidity_accuracy avg_notification_time	id (PK) (FK → test_sessions.id)

Figure 7. *Data Dictionary*

Prototype

Figure 8. *Prototype 1*

Figure 9. *Prototype 2*

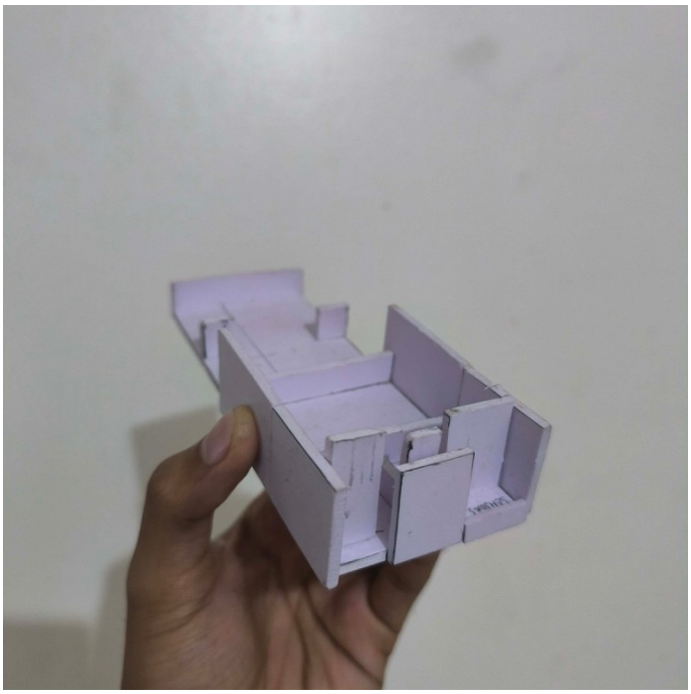


Figure 10. *Prototype 3*

Figure 11. *Prototype 4*

Detail render of the canopy and camera gimbal assembly. The protective canopy measures 20 cm × 55.24 cm. The camera enclosure (housing the sensors) measures 86 mm wide × 177.23 mm long × 80.69 mm tall, mounted on a servo-driven pan bracket for the three checkpoint angles (30°, 90°, 150°).

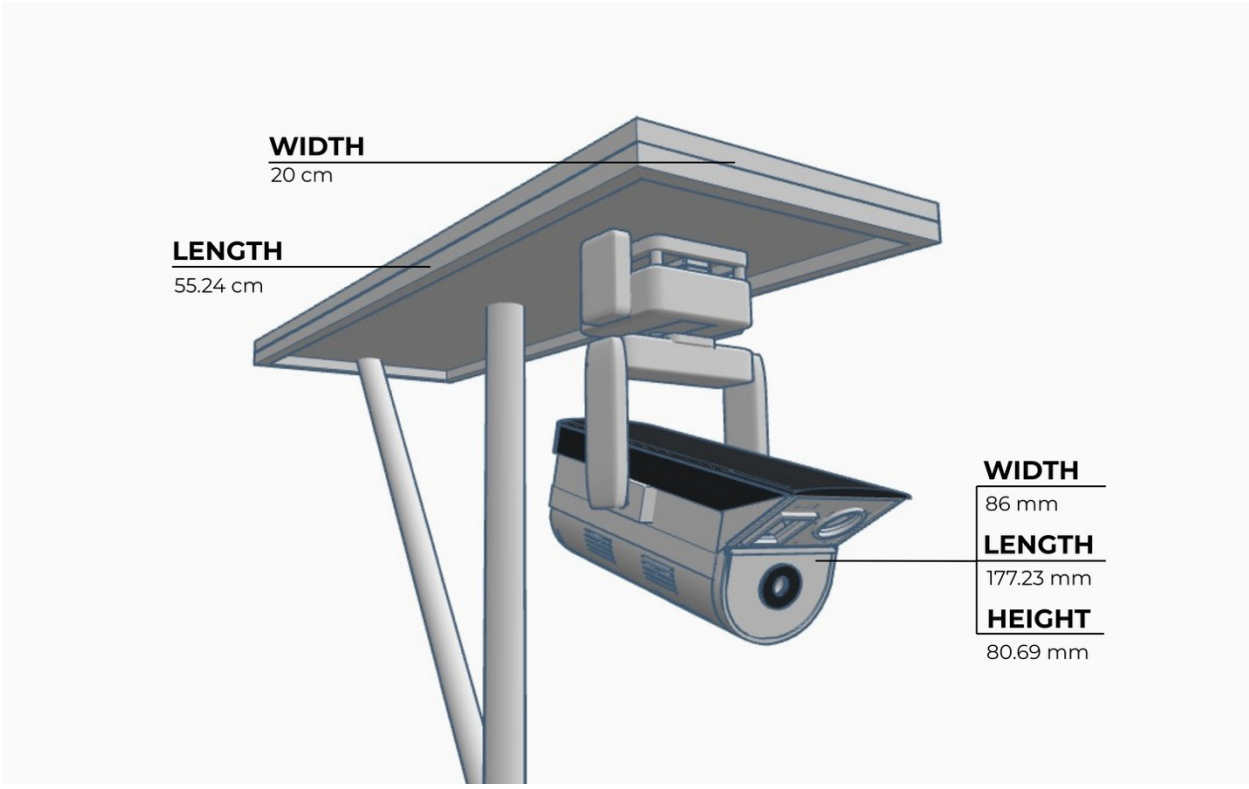


Figure 12. *Final Prototype with Tower (Angle 1)*

Bottom stabilization frame with a square perimeter of 2 ft × 2 ft (all sides) and a diagonal cross-brace length of 20.75 inches. The vertical pole rises 7 feet from the center. The X-brace pattern distributes lateral load to all four corners.

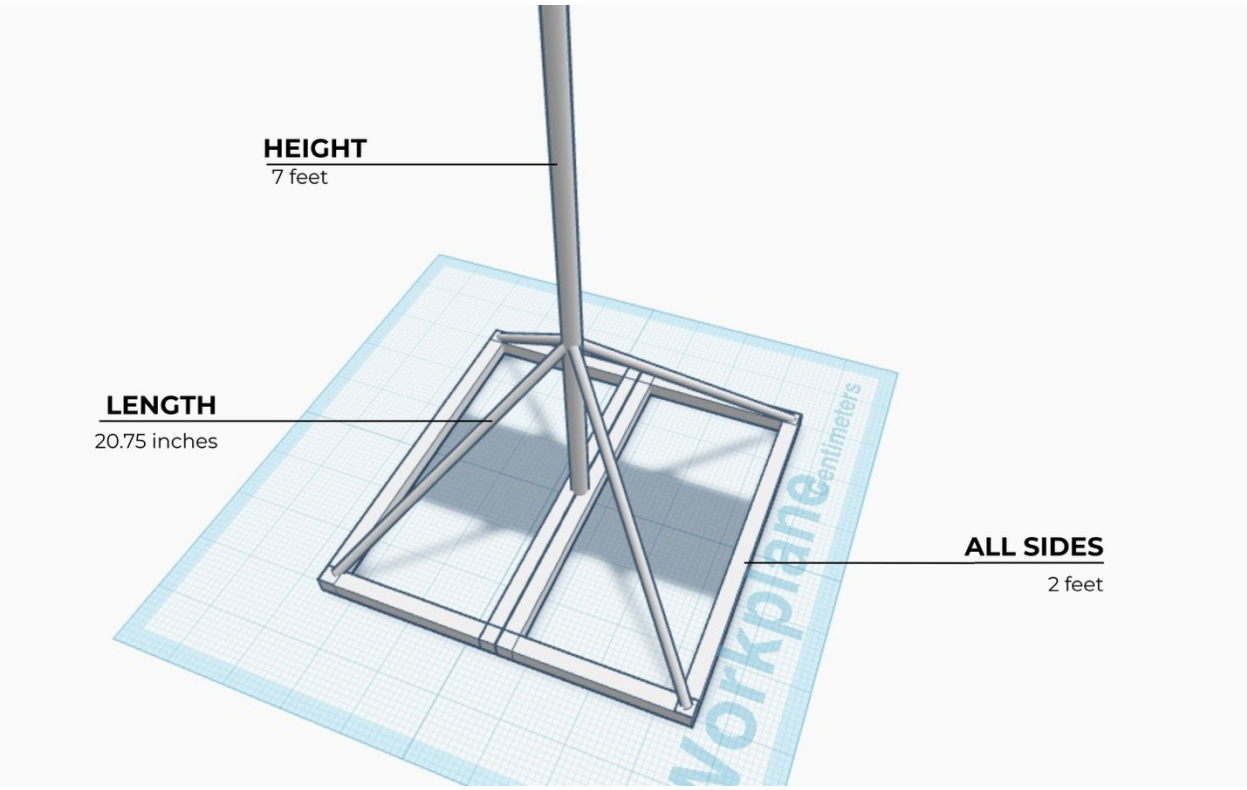


Figure 13. *Final Prototype with Tower (Angle 2)*

Isometric 3D render of the complete FireSafe sensor node tower. A single vertical pole is anchored by a square perimeter base frame with four diagonal braces converging at the pole's base. The camera module is mounted at the top with a canopy/shade panel.

Figure 14. *Final Prototype with Tower (Complete view)*

Hardware Requirements

The following are the components to be used by the system

Component	Functionality
Raspberry Pi Zero 2 W	Serves as the main processor, running TinyML models, handling sensor inputs, and transmitting data to the cloud via its built-in Wi-Fi.
MLX90640	Captures infrared heat signatures for early fire detection, even in smoke or low light.
MQ-2	Detects combustible gases and smoke to support fire-risk confirmation.
ADS1115	Analog-to-digital converter enabling accurate MQ-2 gas readings via I2C.
DHT22	Monitors ambient temperature and humidity to assess fire-prone conditions.
Digital Servo	Controls thermal camera orientation for expanded monitoring coverage.
Arduino Mega 2560 + WiFi	Main processor for SMS Gateway; monitors cloud database and coordinates alert broadcasting. Integrated ESP8266 WiFi enables Supabase communication.
SIM800L GSM Module (×2)	Sends SMS alerts via mobile network; parallel operation enables rapid community broadcasting.
5V Power Supply	Provides stable power for continuous operation in barangay installations.

Table 1. *Hardware Requirements*

Software Requirements

The following are the components to be used by the system

Components	Functionality
Python	Integrates sensors, preprocesses data, and executes detection logic.

TinyML	Runs on the Raspberry Pi to classify fire-like heat patterns efficiently.
Supabase	Provides backend services, user management, and push notification triggers.
React Native	Provides the mobile interface for fire risk levels and real-time alerts.

Table 2. *Software Requirements*

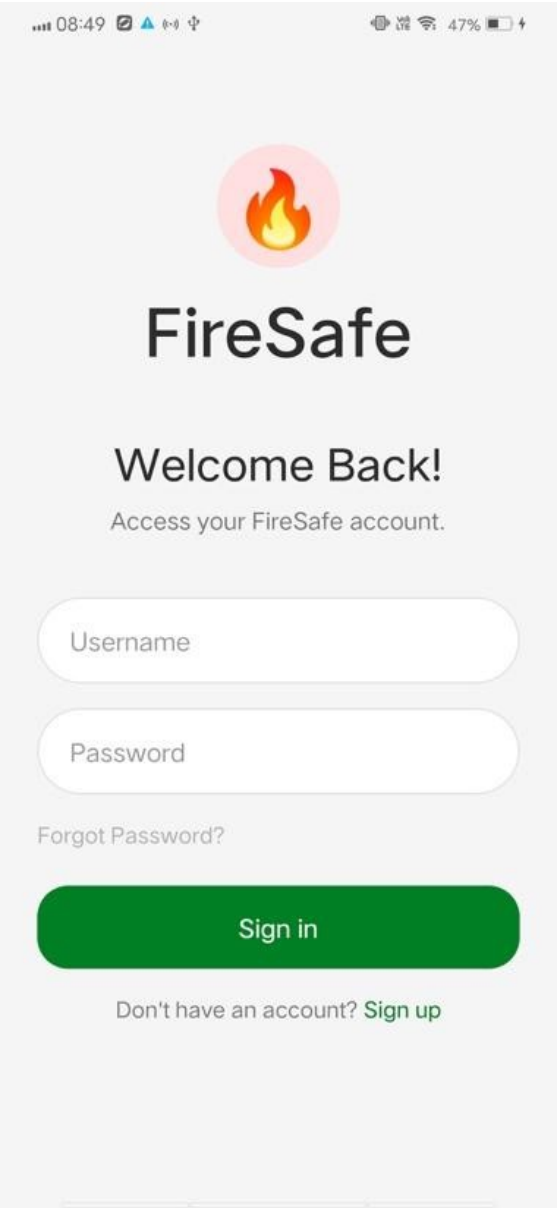
Mobile Application UI Design

Figure 15.

Login Page

Figure 16.

Sign Up Page



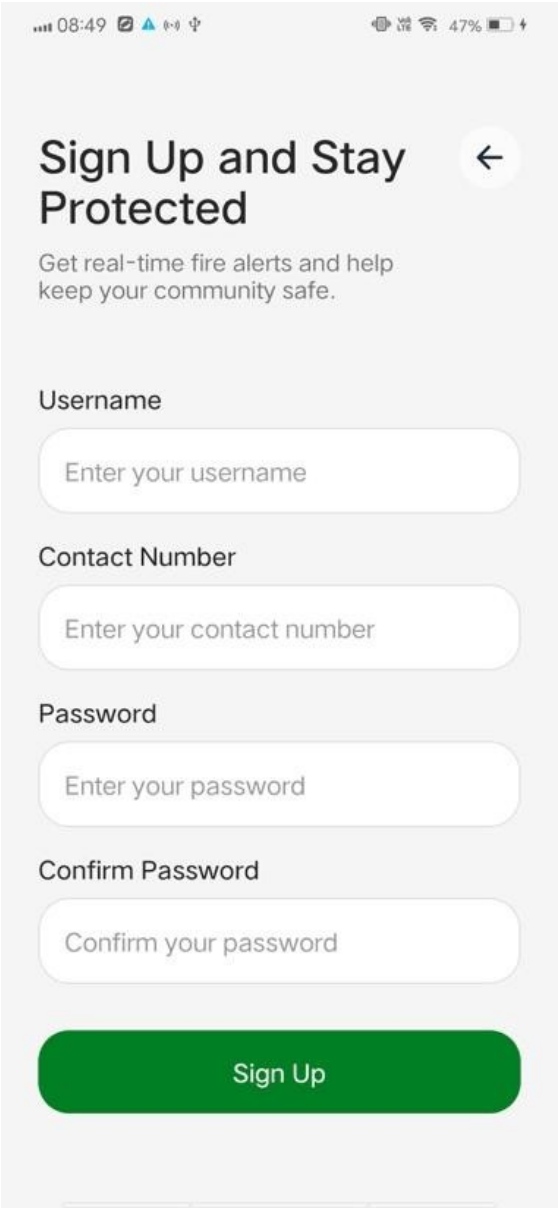


Figure 17.

Dashboard Page

Figure 18.

Node Page

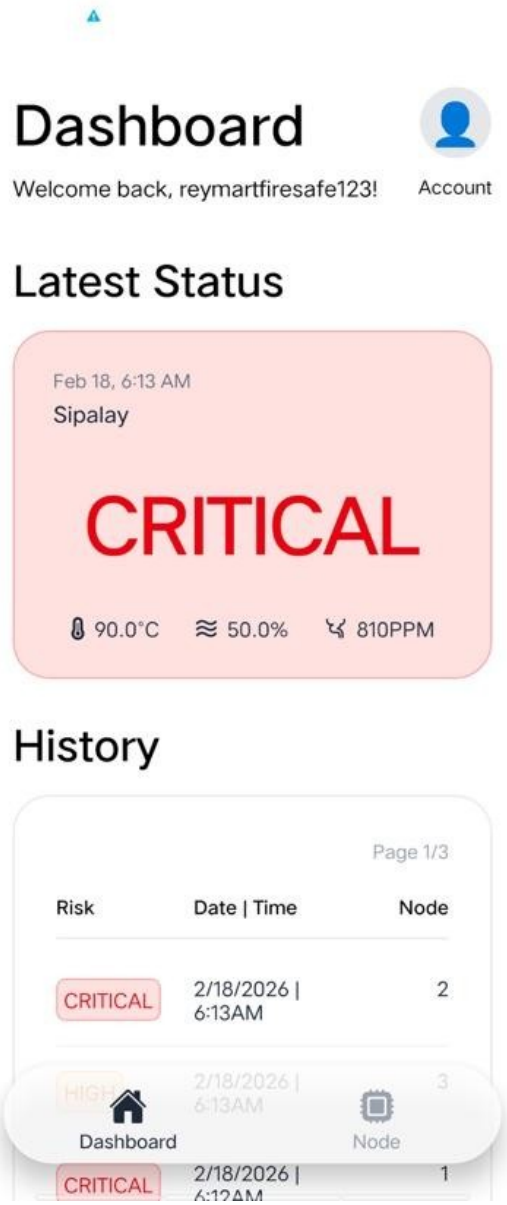


Figure 19.

Specific Node Page

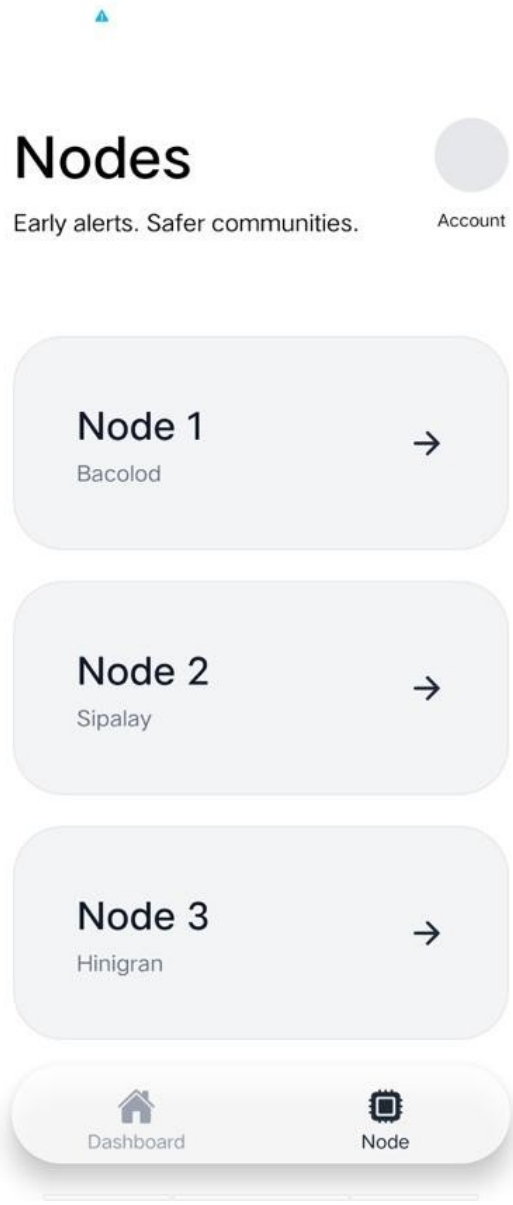


Figure 20.

Nodes Page (Admin)

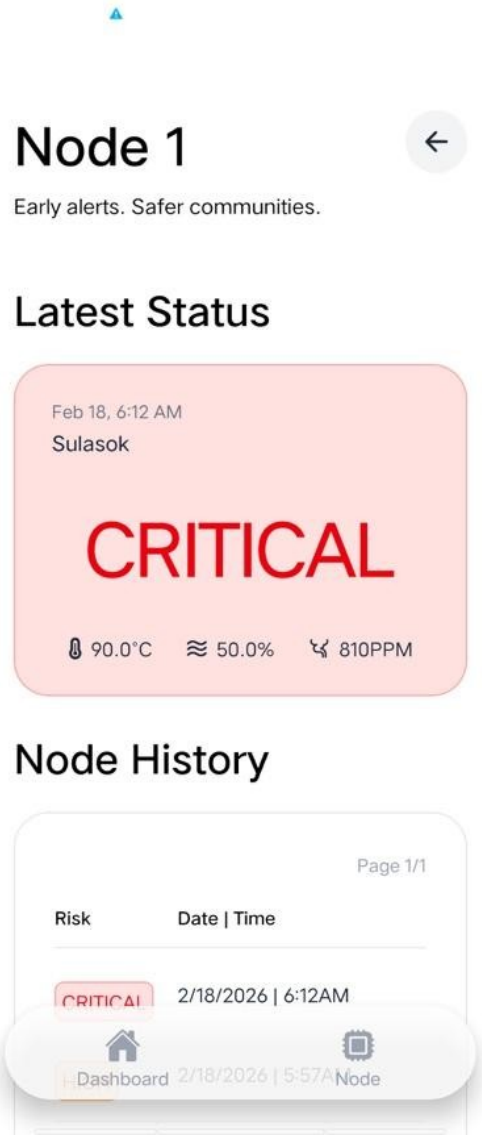


Figure 21.

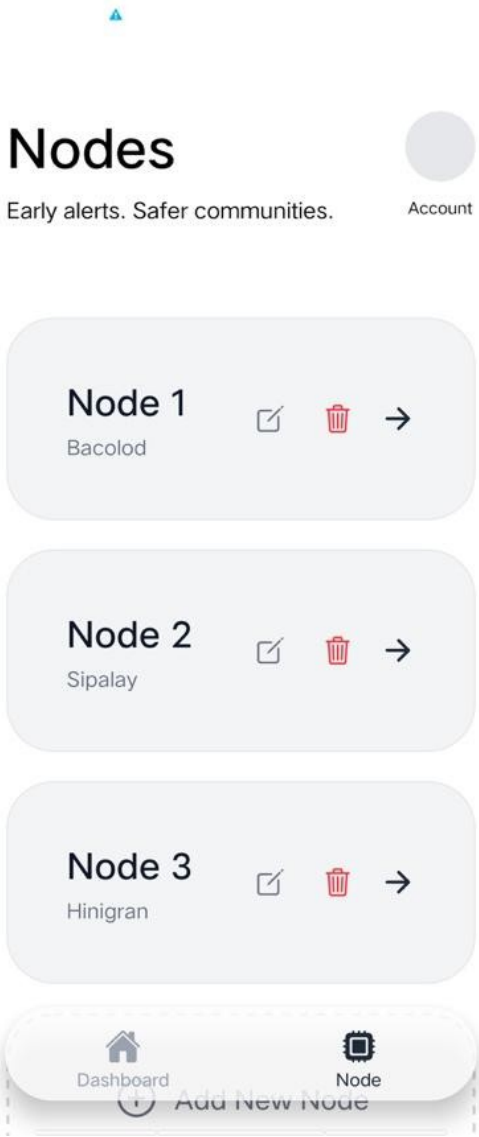


Figure 22.

Add New Node (Admin)

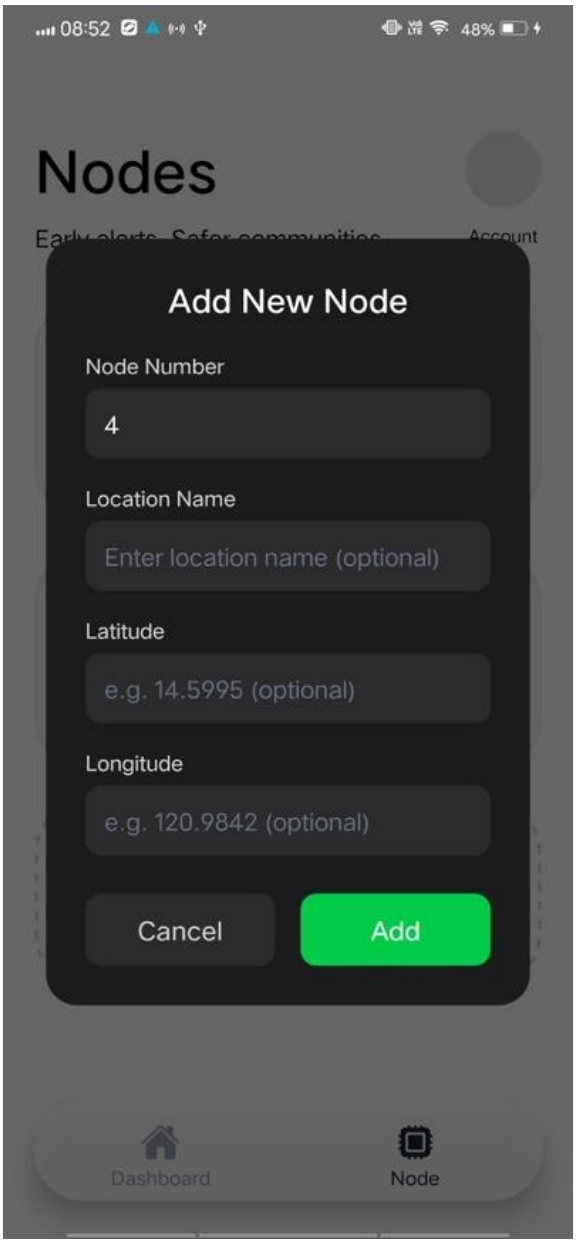


Figure 23.

Edit Node (Admin)

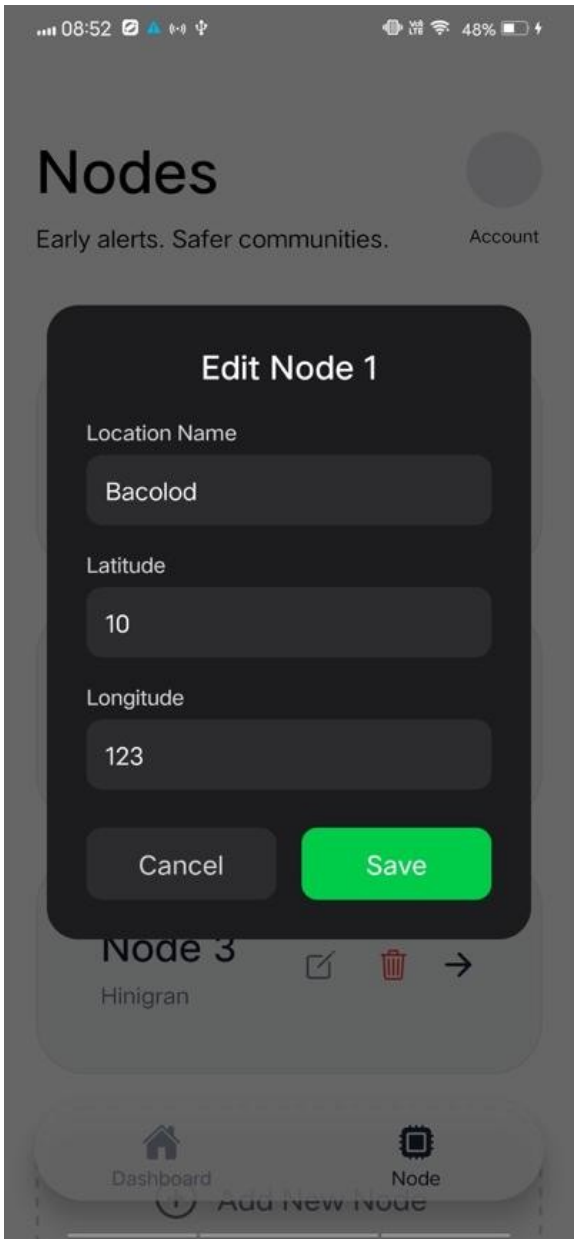
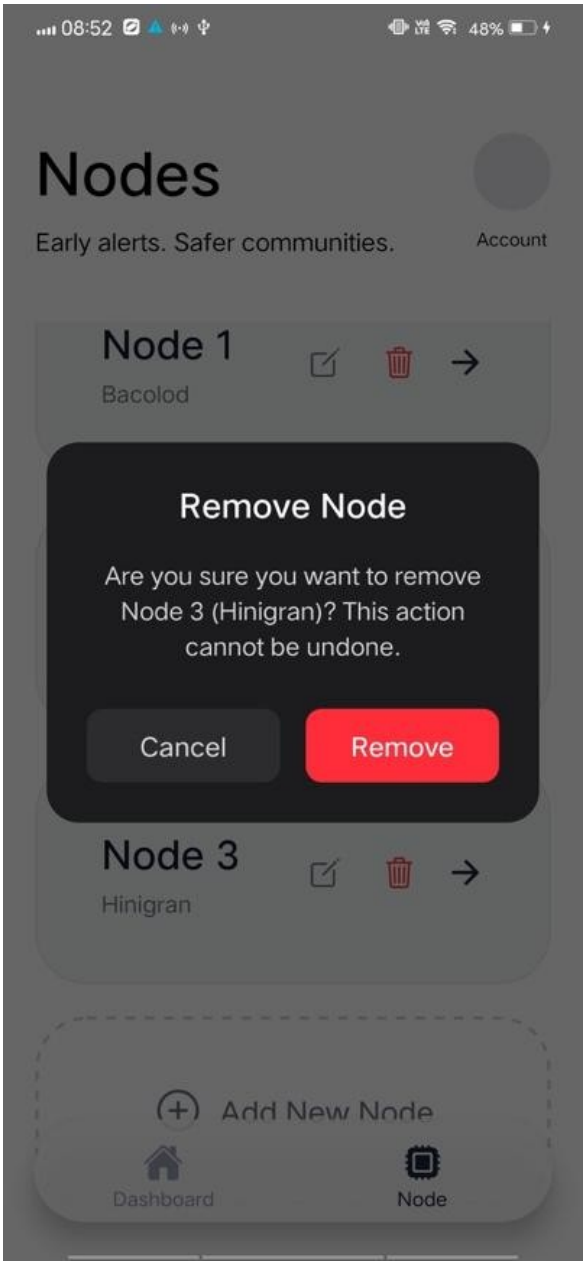
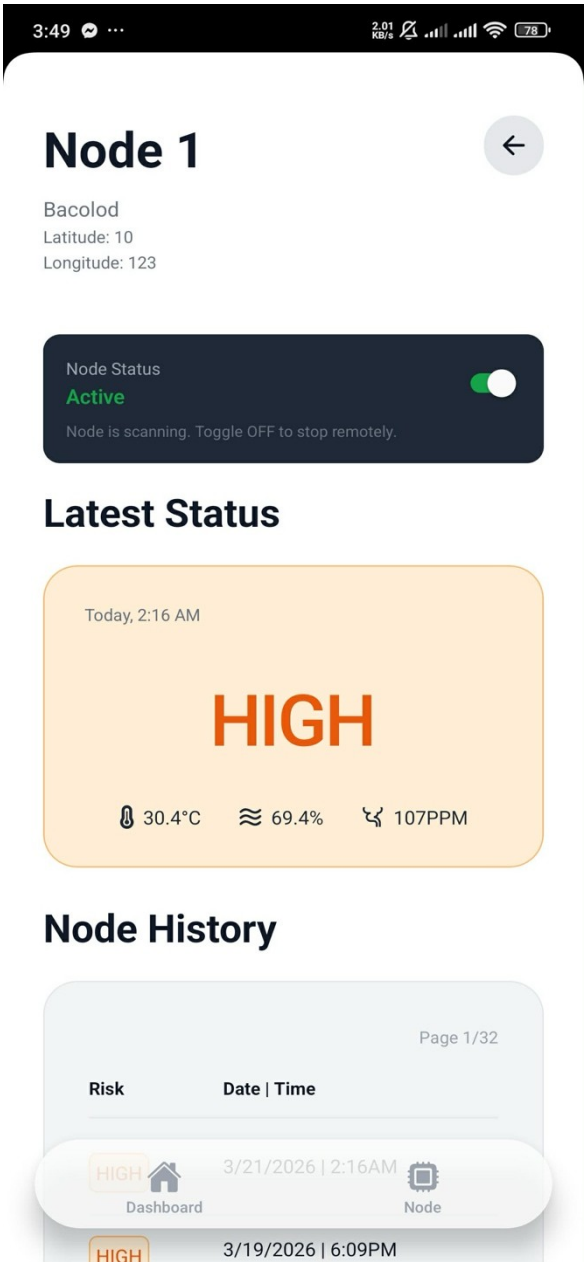


Figure 24.

Remove Node (Admin)



Specific Node Page (Admin)



Data-Gathering Procedure

The data used in this study were gathered automatically through the deployed FireSafe sensor node system, which continuously monitored environmental conditions and logged all fire detection events in real time to a cloud-based Supabase PostgreSQL database. No manual data entry was performed during the testing process. The sensor node, consisting of a Raspberry Pi Zero 2W edge processor, MLX90640 thermal imaging camera, MQ2 gas and smoke sensor, DHT22 temperature-humidity sensor, and ES3 servo motor, processed all sensor inputs locally at the edge, determined whether conditions exceeded fire risk thresholds, and autonomously transmitted the resulting fire event records to the database the moment thresholds were crossed.

Testing was organized into three environments aligned with the study's Statement of the Problem: a controlled indoor environment (Research Question 1), an open space outdoor environment (Research Question 2), and an actual fire scenario (Research Question 3). Each environment consisted of seven independent test sessions, for a total of 21 sessions and 150 individual fire event records across the three environments — 50 per environment. Each session was formally opened by calling the start_test_session() database function, which created a session record containing the environment type, test location, and session start timestamp. At the conclusion of each session, the end_test_session() function was called, which automatically calculated all five performance metrics from the collected fire events and stored the computed results in the session_metrics table.

The system classified each fire event using a multi-sensor threshold logic as defined in Table 1. A fire event was automatically recorded and a push notification was sent to registered devices each time the combination of sensor readings satisfied the HIGH or CRITICAL condition.

Sensor	HIGH Risk Threshold	CRITICAL Risk Threshold
Temperature (DHT22)	≥ 35°C	≥ 38°C
Humidity (DHT22)	≤ 40% RH	≤ 30% RH
Smoke/Gas (MQ2)	100–200 PPM	> 200 PPM

Table 3. *Fire risk classification thresholds applied by the FireSafe system*

Each fire event record automatically captured the following data fields: event timestamp, risk level classification (HIGH or CRITICAL), ambient temperature in degrees Celsius from the DHT22 sensor, smoke gas concentration in PPM from the MQ2 sensor, relative humidity percentage from the DHT22 sensor, servo angle of the MLX90640 camera at the moment of detection, and the session identifier linking the event to its parent session. A notification delivery timestamp was also recorded when the Firebase Cloud Messaging push notification confirmation was received by the backend, allowing notification speed to be computed as the elapsed time in seconds between the event detection timestamp and the delivery confirmation timestamp.

SOP 1: Controlled Indoor Environment

Data for SOP 1 was gathered through seven test sessions conducted at two indoor locations in Eroreco, Kabankalan City: the backyard of Odsey's house on Victoria Street and the backyard of LaSalle Condo. The indoor setting eliminated wind as a confounding variable, allowing smoke to accumulate consistently around the MQ2 sensor, and provided stable WiFi connectivity throughout all sessions.

In each session, the FireSafe sensor node was powered on and left to stabilize for five minutes before a heat stimulus. The system then operated fully autonomously: the MLX90640 thermal camera scanned the area by rotating the servo between the 30, 90, and 150 degree checkpoint positions; the edge processor evaluated incoming sensor readings against the threshold conditions in Table 1; and fire event records were automatically transmitted to the Supabase database each time a threshold was met. The system continued logging events as conditions escalated from HIGH to CRITICAL without any manual intervention. Each of the seven sessions produced between six and eight fire event records, yielding a total of 50 fire events for SOP 1.

Figure 25. *Controlled Environment Test Cases Scenario*

SOP 2: Open Space Outdoor Environment

Data for SOP 2 was gathered through seven test sessions conducted at two outdoor locations in Eroreco, Kabankalan City: the rooftop of a residential building in Barangay 27 and an abandoned field at CentralE. The outdoor setting introduced environmental variables

characteristic of real barangay deployment conditions: wind dispersed smoke away from the MQ2 sensor, reducing gas concentration readings compared to the enclosed indoor sessions; solar heat during daytime sessions elevated the ambient temperature baseline as measured by the DHT22; and mobile hotspot connectivity introduced greater variability in push notification delivery times.

The system operated identically to SOP 1, autonomously scanning, detecting, and logging fire events without manual involvement. No actual fire was started during SOP 2 sessions. Sessions were distributed across morning and afternoon time slots across the six-day testing period to capture the effect of varying ambient conditions. Each of the seven sessions produced between six and eight fire event records, yielding a total of 50 fire events for SOP 2.

Figure 26. *Controlled Environment Test Cases Scenario*

SOP 3: Actual Fire Scenario

Data for SOP 3 was gathered, through seven burn sessions conducted at the CentralE open area in Eroreco, Mandalagan, Bacolod City, 6100 Negros Occidental. The CentralE open area is an abandoned lot with no structures, vegetation of concern, or human occupants in proximity, making it a suitable site for controlled burn testing. Fire extinguishers and containment equipment were on standby throughout all sessions.

Before each burn, combustible materials, dry cogon grass, dried mango leaves, or a combination of grass and leaf material, were arranged at a distance of 2.5 to 4 meters from the sensor tower. The session was actively monitoring before ignition was applied. During the pre-ignition window of two to four minutes, the MLX90640 thermal camera detected heat building up in the prepared fuel pile and the system automatically generated HIGH risk fire event records based on the thermal signature and rising gas concentration, before visible flames appeared. The fuel was then ignited. As active combustion developed, sensor readings escalated and the system logged CRITICAL fire events automatically. The fire was extinguished after the CRITICAL phase was captured. Each of the seven sessions produced between six and eight fire event records, yielding a total of 50 fire events for Research Question 3 and a combined dataset of 150 fire events across all three research environments.

Figure 27. *Controlled Environment Test Cases Scenario*

Statistical Treatment

The statistical treatment of this study addressed SOP 1 through 4 across five performance metrics: (a) detection rate, (b) heat signature detection, (c) gas and smoke detection, (d) ambient temperature and humidity accuracy, and (e) notification speed. The unit of analysis is the individual fire detection event, with N = 50 fire events per environment and a total of N = 150 fire events across three groups. All statistical analyses were performed using IBM SPSS Statistics Version 26.

For Research Questions 1, 2, and 3, descriptive statistics were used to characterize system performance within each environment. These include frequency counts and percentage rates for detection, means and value ranges for sensor readings, and average notification delivery speed. Results are presented in tabular form for each research environment.

For Research Question 4, which asks whether a statistically significant difference exists in system performance across the three environments, the Kruskal-Wallis H-test was employed. The Kruskal-Wallis H-test is a non-parametric alternative to one-way analysis of variance that evaluates differences across three or more independent groups using ranked observations. It does not require the assumption of normally distributed data, making it appropriate for sensor measurement data collected under varying real-world environmental conditions.

Prior to conducting the Kruskal-Wallis test, the Shapiro-Wilk test was applied to assess normality for each metric within each group. Results of the Shapiro-Wilk test yielded $p < 0.05$ for all metrics across all three environments, confirming that the data does not follow a normal distribution and validating the use of a non-parametric statistical approach. A significance level of $\alpha = 0.05$ was applied. A separate Kruskal-Wallis H-test was conducted for each of the five performance metrics, and results are reported as $H(2) = [value]$, $p = [value]$.

Where the Kruskal-Wallis test yielded a statistically significant result ($p < 0.05$), post-hoc pairwise comparisons were performed using the Mann-Whitney U test with Bonferroni correction to identify which specific environment pairs differed significantly. The three pairwise comparisons tested were: controlled environment versus open space, controlled environment versus actual fire, and open space versus actual fire. The Bonferroni correction adjusts the significance threshold for multiple comparisons to $\alpha = 0.017$ (0.05 divided by three comparisons), controlling for family-wise error rate. Adjusted p-values are reported for each pairwise comparison.

RQ	Research Sub-Question	Statistical Treatment	Software
1a–3a	Detection rate in each environment	Descriptive statistics: frequency count and percentage of HIGH and CRITICAL events per environment. Kruskal-Wallis H-test for RQ4a cross-environment comparison.	SPSS v26

1b–3b	Heat signature detection in each environment	Descriptive statistics: mean, minimum, and maximum ambient temperature (°C) per environment. Kruskal-Wallis H-test for RQ4b cross-environment comparison.	SPSS v26
1c–3c	Gas and smoke detection in each environment	Descriptive statistics: mean, minimum, and maximum smoke gas concentration (PPM) per environment. Kruskal-Wallis H-test for RQ4c cross-environment comparison.	SPSS v26
1d–3d	Ambient temperature and humidity accuracy in each environment	Descriptive statistics: percentage of fire events with valid DHT22 sensor readings within operational range per environment. Kruskal-Wallis H-test for RQ4d cross-environment comparison.	SPSS v26
1e–3e	Notification speed in each environment	Descriptive statistics: mean, minimum, and maximum notification delivery time (seconds) per environment. Kruskal-Wallis H-test for RQ4e cross-environment comparison.	SPSS v26
4	Significant difference in performance across controlled environment, open space, and actual fire	Kruskal-Wallis H-test (non-parametric, $\alpha = 0.05$, df = 2). Post-hoc pairwise Mann-Whitney U test with Bonferroni correction (adjusted $\alpha = 0.017$) for significant results. Normality verified using Shapiro-Wilk test.	SPSS v26

Table 4. *Statistical treatment applied to each research question and performance metric*

RESULTS AND DISCUSSIONS

Results

This section presents the performance outcomes of the FireSafe automated fire detection system across three environments. A total of 150 fire events were recorded across 21 test sessions with 50 events per environment. Table 1 summarizes the key descriptive metrics.

Environment	N	Mean Temp (°C)	Mean Smoke (PPM)	Mean Humidity (%)	Avg Notif Speed (s)	Detection Rate (%)	Temp/Humidity Accuracy (%)
Controlled Indoor	50	34.51	72.34	55.89	2.41	100.00	97.14
Open Space	50	41.34	480.29	42.71	4.32	100.00	100.00

Outdoor							
Actual Fire	50	27.45	167.89	81.06	4.18	100.00	88.57

Table 5. *Descriptive summary of FireSafe system performance (N = 50 per environment)*

SOP 1: Controlled Indoor Environment

The system achieved a 100% detection rate across all 50 fire events in the controlled indoor environment. Ambient temperature readings ranged from 33.5°C to 35.0°C (mean = 34.51°C), and all events that met the HIGH threshold (≥ 35°C) were correctly flagged. Smoke gas readings ranged from 13.7 to 228.5 PPM (mean = 72.34 PPM), with HIGH (100–200 PPM) and CRITICAL (> 200 PPM) threshold crossings detected in sessions where the heat stimulus was introduced. Humidity averaged 55.89% relative humidity. The ambient temperature and humidity accuracy rate was 97.14% — one event in Session 1 returned a reading slightly outside the DHT22 valid range during sensor warm-up. Average notification speed was 2.41 seconds, reflecting stable indoor WiFi connectivity throughout all sessions.

SOP 2: Open Space Outdoor Environment

The system achieved a 100% detection rate across all 50 fire events in the open space outdoor environment. Ambient temperature readings ranged from 35.7°C to 50.6°C (mean = 41.34°C), elevated by solar exposure during daytime sessions. All 50 events exceeded the HIGH temperature threshold (≥ 35°C). Smoke gas readings ranged from 305.0 to 651.0 PPM (mean = 480.29 PPM), all exceeding the CRITICAL threshold (> 200 PPM), reflecting higher ambient gas concentrations in the outdoor field environment. Humidity averaged 42.71% relative humidity. Temperature and humidity accuracy was 100% across all events. Average notification speed was 4.32 seconds, reflecting variable mobile hotspot connectivity at the outdoor test locations.

SOP 3: Actual Fire Scenario

The system achieved a 100% detection rate across all 50 fire events in the actual fire scenario. The DHT22 recorded ambient temperatures ranging from 25.2°C to 30.3°C (mean = 27.45°C), lower than the other environments due to early morning testing (1:30–6:00 AM). HIGH risk events were recorded two to four minutes before ignition, as the MLX90640 thermal camera detected heat signatures in the fuel pile before visible flames appeared. Smoke gas readings ranged from 103.0 to 271.0 PPM (mean = 167.89 PPM). Humidity averaged 81.06% relative humidity, reflecting early morning atmospheric moisture and water vapor from combustion. Temperature and humidity accuracy was 88.57%, as some CRITICAL-phase events produced ambient readings affected by active fire conditions. Average notification speed was 4.18 seconds.

SOP 4: Statistical Comparison Across Environments

Normality Test

The Shapiro-Wilk test was applied to all metrics across all three groups prior to statistical testing. All results yielded $p < 0.05$, confirming non-normal distribution and validating the use of the Kruskal-Wallis H-test.

Tests of Normality

	Group	Kolmogorov-Smirnov[a] Statistic	df	Sig.	Shapiro-Wilk Statistic	df	Sig.
Temperature_C	Control	.127	50	.043	.924	50	.003
	Open Space	.177	50	.000	.908	50	.001
	Actual Fire	.390	50	.000	.651	50	.000
Smoke_Gas_PPM	Control	.471	50	.000	.398	50	.000
	Open Space	.196	50	.000	.862	50	.000
	Actual Fire	.382	50	.000	.677	50	.000
Humidity_Pct	Control	.173	50	.001	.909	50	.001
	Open Space	.134	50	.025	.934	50	.008
	Actual Fire	.383	50	.000	.650	50	.000
Notif_Speed_s	Control	.324	50	.000	.747	50	.000
	Open Space	.217	50	.000	.843	50	.000
	Actual Fire	.175	50	.001	.920	50	.002

a. Lilliefors Significance Correction

Figure 28. *Shapiro-Wilk and Kolmogorov-Smirnov tests of normality (all metrics, all environments)*

Kruskal-Wallis H-Test

Table 2 presents the Kruskal-Wallis H-test results for all four sensor metrics at $\alpha = 0.05$.

Metric	H Statistic	df	p-value	Decision
Temperature (°C)	133.572	2	0.000	Reject H_0 — Significant difference
Smoke Gas (PPM)	125.361	2	0.000	Reject H_0 — Significant difference
Humidity (%)	117.887	2	0.000	Reject H_0 — Significant difference
Notification Speed (s)	104.517	2	0.000	Reject H_0 — Significant difference

Table 6. *Kruskal-Wallis H-test results across three environments*

All four metrics yielded statistically significant results: temperature $H(2) = 133.572$, $p < 0.001$; smoke gas $H(2) = 125.361$, $p < 0.001$; humidity $H(2) = 117.887$, $p < 0.001$; notification speed $H(2) = 104.517$, $p < 0.001$. The null hypothesis was rejected for all four metrics.

Post-Hoc Pairwise Comparisons

Mann-Whitney U tests with Bonferroni correction (adjusted $\alpha = 0.017$) were conducted for all significant metrics. Table 3 presents the complete results.

Metric	Group 1	Group 2	Test Statistic	Adj. p-value	Sig.
Temperature	Controlled	Open Space	-50.000	0.000	Yes
Temperature	Controlled	Actual Fire	50.000	0.000	Yes
Temperature	Open Space	Actual Fire	100.000	0.000	Yes
Smoke Gas	Controlled	Actual Fire	-44.000	0.000	Yes
Smoke Gas	Controlled	Open Space	-97.000	0.000	Yes
Smoke Gas	Actual Fire	Open Space	53.000	0.000	Yes
Humidity	Open Space	Controlled	37.040	0.000	Yes
Humidity	Open Space	Actual Fire	-93.520	0.000	Yes
Humidity	Controlled	Actual Fire	-56.480	0.000	Yes
Notification Speed	Controlled	Actual Fire	-66.800	0.000	Yes
Notification Speed	Controlled	Open Space	-83.200	0.000	Yes
Notification Speed	Actual Fire	Open Space	16.400	0.171	No

Table 3. *Post-hoc pairwise Mann-Whitney U test results with Bonferroni correction*

For temperature, smoke gas, and humidity, all three environment pairs yielded significant pairwise differences (Adj.p = 0.000). For notification speed, two of three pairs were significant, controlled versus actual fire (Adj.p = 0.000) and controlled versus open space (Adj.p = 0.000), while the comparison between actual fire and open space was not significant (Adj.p = 0.171).

Each node shows the sample average rank of Group.

Sample1-Sample2	Test Statistic	Std. Error	Std. Test Statistic	Sig.	Adj.Sig.
Actual Fire-Control	50.000	8.653	5.779	.000	.000
Actual Fire-Open Space	100.000	8.653	11.557	.000	.000
Control-Open Space	-50.000	8.653	-5.779	.000	.000

Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same.
Asymptotic significances (2-sided tests) are displayed. The significance level is .05.
Significance values have been adjusted by the Bonferroni correction for multiple tests.

Figure 29. *Pairwise comparison for temperature.*

Each node shows the sample average rank of Group.

Sample1-Sample2	Test Statistic	Std. Error	Std. Test Statistic	Sig.	Adj.Sig.
Control-Actual Fire	-44.000	8.676	-5.072	.000	.000
Control-Open Space	-97.000	8.676	-11.180	.000	.000
Actual Fire-Open Space	53.000	8.676	6.109	.000	.000

Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same.
Asymptotic significances (2-sided tests) are displayed. The significance level is .05.
Significance values have been adjusted by the Bonferroni correction for multiple tests.

Figure 30. *Pairwise comparison for smoke gas*

Each node shows the sample average rank of Group.

Sample1-Sample2	Test Statistic	Std. Error	Std. Test Statistic	Sig.	Adj.Sig.
Open Space-Control	37.040	8.675	4.270	.000	.000
Open Space-Actual Fire	-93.520	8.675	-10.780	.000	.000
Control-Actual Fire	-56.480	8.675	-6.511	.000	.000

Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same.
Asymptotic significances (2-sided tests) are displayed. The significance level is .05.
Significance values have been adjusted by the Bonferroni correction for multiple tests.

Figure 31. *Pairwise comparison for humidity*

Each node shows the sample average rank of Group.

Sample1-Sample2	Test Statistic	Std. Error	Std. Test Statistic	Sig.	Adj.Sig.
Control-Actual Fire	-66.800	8.622	-7.748	.000	.000
Control-Open Space	-83.200	8.622	-9.650	.000	.000
Actual Fire-Open Space	16.400	8.622	1.902	.057	.171

Each row tests the null hypothesis that the Sample 1 and Sample 2 distributions are the same.
Asymptotic significances (2-sided tests) are displayed. The significance level is .05.
Significance values have been adjusted by the Bonferroni correction for multiple tests.

Figure 32. *Pairwise comparison for notification speed.*

Discussion

This section interprets the results in relation to the study's research questions and explains the observed patterns in the context of the FireSafe system's design and deployment environment.

Detection Rate Across All Environments

The FireSafe system achieved a 100% detection rate across all 150 fire events in all three environments. This consistent result confirms that the multi-sensor fusion approach, combining the MLX90640 thermal camera, MQ2 gas sensor, and DHT22 temperature-humidity sensor, is sufficient to reliably detect fire risk conditions regardless of whether the setting is a controlled indoor room, an open outdoor field, or an actual fire scenario. The 100% detection rate is particularly significant for the open space environment, where wind dispersal would have reduced the effectiveness of a smoke-only detection system. The thermal camera compensated for reduced smoke gas concentration by providing heat signature data that independently satisfied the detection threshold. This finding directly addresses the study's central argument for using thermal imaging in rural barangay deployments.

Pre-Ignition Early Warning Detection

The most significant finding of this study is the pre-ignition detection demonstrated in the actual fire sessions. The MLX90640 thermal camera detected heat building up in the fuel pile two to four minutes before ignition, automatically generating HIGH risk alerts before visible flames appeared. This capability, detecting fire risk through heat signature analysis prior to active combustion, is not achievable with conventional smoke detectors or simple gas sensors, which can only respond after combustion has begun. This result directly answers the study's problem statement: rural barangay communities currently lack any automated monitoring mechanism

capable of providing early warning before a fire becomes uncontrollable. The FireSafe system demonstrates this capability under actual field conditions.

Environmental Influence on Sensor Readings

The Kruskal-Wallis test confirmed statistically significant differences across all four sensor metrics ($p < 0.001$ for all), indicating that the three environments produced measurably distinct sensor behavior profiles. However, these differences reflect the physical characteristics of each environment rather than system failures. The open space environment produced the highest temperature and smoke gas readings due to solar exposure and ambient outdoor gas concentrations. The actual fire environment produced the highest humidity due to early morning atmospheric moisture and combustion-generated water vapor, and the lowest ambient temperature due to early morning testing. These patterns are consistent with the known physical conditions of each environment and support the validity of the data collected.

Notification Speed and Network Dependency

The controlled indoor environment produced significantly faster notification delivery (mean = 2.41 s) compared to both outdoor environments (4.32 s and 4.18 s), as confirmed by post-hoc pairwise comparison (Adj.p = 0.000 for both pairs). However, the comparison between the open space and actual fire environments was not statistically significant (Adj.p = 0.171), indicating that both outdoor settings produced comparable notification latency. This finding reveals that notification speed is primarily determined by network connectivity quality, indoor WiFi versus outdoor mobile hotspot, rather than by the nature or severity of the fire event. The practical implication for barangay deployment is that improving local wireless infrastructure at sensor node locations would be the most effective intervention for reducing alert delivery time in outdoor settings.

Temperature and Humidity Accuracy

The controlled environment accuracy rate of 97.14% and the actual fire environment rate of 88.57% reflect the known operational limits of the DHT22 sensor. In the controlled environment, one warm-up reading during Session 1 produced an out-of-range value before the sensor stabilized, a minor and expected behavior during initial power-on. In the actual fire environment, some CRITICAL-phase events produced ambient thermal and moisture conditions that approached or slightly affected DHT22 readings. These accuracy reductions do not compromise the system's fire detection capability, as the MLX90640 thermal camera operates independently of DHT22 and continued to detect heat signatures accurately throughout all sessions. The DHT22 serves as a supplementary environmental context sensor, while the thermal camera is the primary detection modality.

Overall System Reliability

Across all 21 sessions and 150 fire events, the FireSafe system delivered a push notification for every detected event without a single missed alert. The consistent 100% detection rate, combined with sub-5-second notification delivery across all environments, demonstrates that the system is capable of reliable real-time fire risk monitoring under the range of conditions representative of rural barangay deployment. The statistically significant differences in sensor readings across environments do not diminish this conclusion, they instead reflect the environmental diversity that any deployed system must contend with, and confirm that the multi-sensor architecture is sufficiently robust to maintain detection reliability across that diversity.

REFERENCES

Almeida, J. S., Huang, C., Nogueira, F. G., Bhatia, S., & Albuquerque, de Victor Hugo C. (2022). EdgeFireSmoke: A Novel Lightweight CNN Model for Real-Time Video Fire–Smoke Detection. *IEEE Transactions on Industrial Informatics*, 18(11), 7889-7898.

https://ieeexplore.ieee.org/abstract/document/9684982

Ateeq, Z., & Momani, M. (2020). Wireless Sensor Networks using Image Processing for Fire Detection. *5th International Conference on Innovative Technologies in Intelligent Systems and Industrial Applications (CITISIA)*, 1-10.

https://ieeexplore.ieee.org/document/9371798

Bureau of Fire Protection (2024). CY 2023 Annual Narrative Accomplishment Report. Quezon City: BFP.

https://bfp.gov.ph/wp-content/uploads/2024/02/CY-2023-Annual-Narrative-Accomplishment-Report.pdf

Dampage, U., Bandaranayake, L., Wanasinghe, R., Kottahachchi, K., & Jayasanka, B. (2022). Forest Fire Detection System using Wireless Sensor Networks and Machine Learning. *Scientific Reports*, 12(46).

https://doi.org/10.1038/s41598-021-03882-9

Das, S., Das, J., Krishna, O.B., & Maiti J. (2022). Image Processing-Based Fire Detection Using IoT Devices. *CRC Press*¸ 18.

https://doi.org/10.1201/9781003122401-10

Dhir, K., Singh, P., Das, R., & Sawhney, R. S. (2023). Fusing Innovation and Nature: Empowering Forest Fire Detection and Prediction through IoT enabled Edge Computing and Deep Learning. *IFIP Advances in Information and Communication Technology*, 699, 225-236.

https://doi.org/10.1007/978-3-031-50204-0_19

Ferreira, L. M., Coimbra, A. P., de Almeida, A. T. (2020). Autonomous System for Wildfire and Forest Fire Early Detection and Control. *Inventions*, *5*(3), 41.

https://doi.org/10.3390/inventions5030041

Guo, C., Bai, Y., Wu, M., & Zhou, Y. (2022). Real-Time Fire Detection and Alarm System using Edge Computing and Cloud IoT Platform. *International Journal of Wireless and Mobile Computing*, 22(3-4), 310-318.

https://doi.org/10.1504/ijwmc.2022.124822

Komalapati, N., Yarra, V. C., Kancharla, L. A. V., & Shankar, T. N. (2021). Smart Fire Detection and Surveillance System Using IoT. *2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS)*, 1386-1390.

https://ieeexplore.ieee.org/document/9395841

Mahdi, A. S., & Mahmood, S. A. (2022). An Edge Computing Environment for Early Wildfire Detection. *Annals of Emerging Technologies in Computing*, 6(3), 56–68.

https://doi.org/10.33166/aetic.2022.03.005

Mekni, K. (2022). Design and Implementation of a Smart Fire Detection and Monitoring System based on IoT. *2022 4th International Conference on Applied Automation and Industrial Diagnostics (ICAAID)*, 1-5.

https://ieeexplore.ieee.org/document/9799505

Mohammed, R. K. (2022). A Real-Time Forest Fire and Smoke Detection System using Deep Learning. *International Journal of Nonlinear Analysis and Applications*, 13(1). https://ijnaa.semnan.ac.ir/article_5899.html

Mu, L., Yang, Y., Wang, B., Zhang, Y., Feng, N., & Xie, X. (2025). Edge Computing-Based Real Time Forest Fire Detection Using UAV Thermal and Color Images. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 18, 6760-6771. https://ieeexplore.ieee.org/abstract/document/10838709

Sousa, M. J., Moutinho, A., & Almeida, M. (2020). Thermal Infrared Sensing for Near Real Time Data-Driven Fire Detection and Monitoring Systems. *Sensors*, 20(23), 6803. https://doi.org/10.3390/s20236803

Su, L., Lee, Y., Chen, Y., Tseng, H., & Yang, C. (2023). Using Edge Computing Technology in Programmable Logic Controller to realize the Intelligent System of Industrial Safety and Fire Protection. *Sensors and Materials*, 35(5), 1731. https://doi.org/10.18494/sam4323

Tsai, P.-F., Liao, C.-H., & Yuan, S.-M. (2022). Using Deep Learning with Thermal Imaging for Human Detection in Heavy Smoke Scenarios. *Sensors*, 22(14), 5351. https://doi.org/10.3390/s22145351

Vazquez, G., Zhai, S., & Yang, M. (2025). Optimized Lightweight YOLOv5 Model for Efficient Wildfire Flame and Smoke Detection on Edge Computing Devices. *IEEE Conference on Technologies for Sustainability (SusTech)*, 1-8. https://ieeexplore.ieee.org/abstract/document/11025794

Vuong, D., Tran, V. V., Tran, V. T., Nhat Tran, H. A., Hao Le, P. H., Son, T. T. (2024). Experimental Fire Detection and Alarm System Based on the Integration of IoT and Image Processing. *2024 IEEE International Conference on Consumer Electronics-Asia (ICCE-Asia)*, 1-4.

https://ieeexplore.ieee.org/document/10773659

Wenzheng, L., & Jie, W. (2023). A YOLOv7 Forest Fire Detection System with Edge Computing. *IEEE 13th International Conference on Electronics Information and Emergency Communication (ICEIEC)*, 223-227.

https://ieeexplore.ieee.org/abstract/document/10200044

Yang, W., Wu, Y. & Chow, S.K.K. (2024). Deep Learning Method for Real-Time Fire Detection System for Urban Fire Monitoring and Control. *Int J Comput Intell Syst*, 17(216).

https://doi.org/10.1007/s44196-024-00592-8

Yang, X., Li, Y., & Chen, Q. (2024). Automated Image-Based Fire Detection and Alarm System Using Edge Computing and Cloud-Based Platform. *Internet or Things*, 28, 101402.

https://doi.org/10.1016/j.iot.2024.101402